Research on Fitting Strategy in HPPC Test for Li-ion battery

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***Abstract*—The objective of this work is to establish an accurate battery model (BM) by hybrid pulse power characteristic (HPPC) test according to state of charge (SOC), temperature and current. The BM is developed for Li-ion battery based on the electrical equivalent circuit (EEC). The influence of different order of EEC and the existing issues in the fitting process are analyzed. Based on these, considering the accuracy and simplicity of the model, a second order EEC model with additional resistance is established and a fitting strategy based on genetic algorithm (GA) by enhancing initial and end point are proposed, which can adjust the order of EEC model according to the fitting accuracy. The result shows that the proposed model has higher accuracy than the traditional model, and this strategy can simplify the complexity of the model and increase the fitting accuracy.**

Keywords—battery model, hybrid pulse power characteristic, genetic algorithm, electrical equivalent circuit, Li-ion.

I. INTRODUCTION

Model plays an important role in researching the performance, charging strategy and vehicle to grid (V2G) control strategy of the battery. Several types of battery models are reported in literatures. Such as electrochemical model, AC impedance spectrum, mathematical and electrical equivalent circuit (EEC) and so on [1-3].

For the reason that there are complex parameters and require extensive experimentation in order to determine the parameters. Electrochemical models are not well suitable for practical application and simulation of the battery. In addition, it cannot fully reflect electrical characteristics and the complexity of chemical reactions making it more difficult to express battery performance under changes of temperature, state of charge (SOC) and current [4]. The mathematical model of the battery is based on the empirical equations, which is limited in the use of circuit simulators as it is not sufficiently related to the model parameters and the current-voltage (I-V) characteristics of the battery [5].

The EEC based models can efficiently represent the characteristics of Li-ion battery, which consists of an open circuit voltage (OCV) changing with SOC, and a series of resistances and parallels combination of resistance and capacitance. These electrical component parameters are obtained by hybrid pulse power characteristic (HPPC) test. I-V characteristics of the battery can be obtained by solving differential equations [6-7].

In order to use the EEC model effectively in charging system, it is important to determine the parameters accurately. The key to ensure the accuracy of the parameters is the accuracy of fitting with HPPC curve. Based on the accurate measurement of OCV and capacity, the characteristic curve of battery under this condition can be obtained by designing reasonable HPPC test. The parameters of EEC model can be obtained by response equation [8-9].

The design of EEC model has a direct impact on the fitting accuracy. as the characteristics of battery vary with temperature, SOC and current, there are conditions where the model order needs to be increased to ensure accuracy. [10-11]. Therefore, a reasonable fitting strategy is needed, which can timely adjust the order of model, otherwise the unstable conditions may occur, such as negative capacitance, resistance or the fitting result is inaccurate.

In addition to nonlinear least squares (NLS), random algorithms such as particle swarm optimization (PSO) and genetic algorithms (GA) can also be used to fit the curves of battery model by designing a reasonable objective function [12-13].

This paper is organized as follows: Section II introduces the design of HPPC test and the fitting process of traditional second-order EEC model. In section III, the issue of fitting results is analyzed. Based on this, a second-order EEC model with additional resistance is proposed to increase accuracy. In section IV, a parameters fitting strategy that can adjust the number order according to the fitting accuracy based on GA is designed and the result is analyzed. Section V concludes this paper.

II. SECOND-ORDER EEC MODEL

A. Design of HPPC and EEC Model

The second-order EEC model is shown in Fig.1. The voltage source represents the open circuit voltage (OCV) of the battery, R_0 represents the DC impedance, and the two sets of parallel *R, C* represent the polarization.

Standard HPPC experiments use 10 s pulses and 40 s stand at each SOC point. In order to observe the fitting effect with different standing time and to ensure the stability of polarization, 30 s pulse and 600 s stand are used here. In addition, a negative pulse of 30 s is added to the front to ensure the accuracy of the SOC point. Currents from 0.5 to 3.0 A are used in the experiment and the temperature is fixed at 20 ºC. HPPC experimental result is shown in Fig. 2.

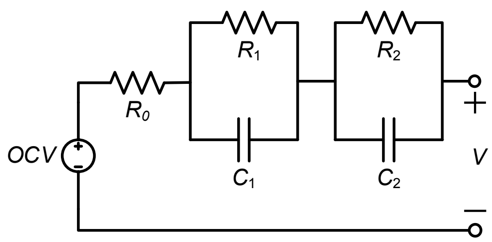

Fig. 1. Second-order EEC model.

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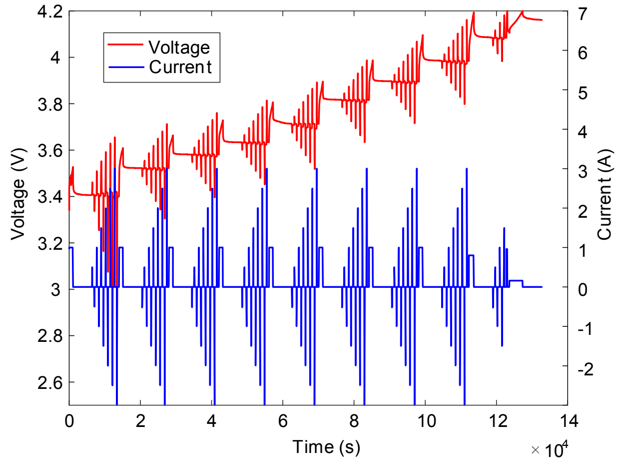

Fig. 2. Result of HPPC experiment.

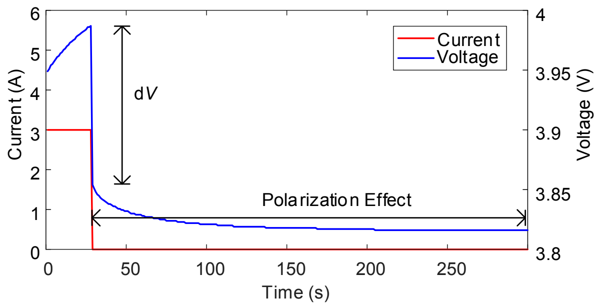

Fig. 3. Battery parameters identification.

B. Process of Fitting

Fig. 3 shows a segment of the pulse experiment. When the charging current drops to zero, the voltage of the battery instantly drops. As the voltage of polarized capacitor cannot suddenly be changed, this voltage drop is the voltage on the DC resistance. Therefore, dV/dI can stand for the DC resistance. During the open cell stage, the energy stored on the polarized capacitor is released by the polarization resistance. Since there is no current input, this process can be regarded as a zero input response. Combined with (1), we can get the equation of voltage change after open cell:

$$V = OCV + V_{10}e^{-t/\tau_1} + V_{20}e^{-t/\tau_2} \tag{1}$$

where τ_1=R_1×C_1, τ_2=R_2×C_2. According to the Fig. 2, U_{10}, U_{20}, τ_1, τ_2 can be obtained by fitting the polarization effects part of the curve. Assuming that the battery state is stable before open cell and the current flows through the polarization resistance, R_1, R_2 can be calculated according to the charging current. Then C_1=τ_1/R_1, C_2=τ_2/R_2. OCV is represented by the terminal voltage after standing for 5000 s after charging to a SOC.

As shown in Fig. 4, the ideal fitting effect should have a good fitting effect in initial stage, arc stage and stable stage. Any inaccuracy in one of the stage will affect the parameters of EEC model. Especially the initial stage and the stable stage, their inaccuracy will cause the decrease of polarization resistance, resulting in the downward shift of the entire fitting curve.

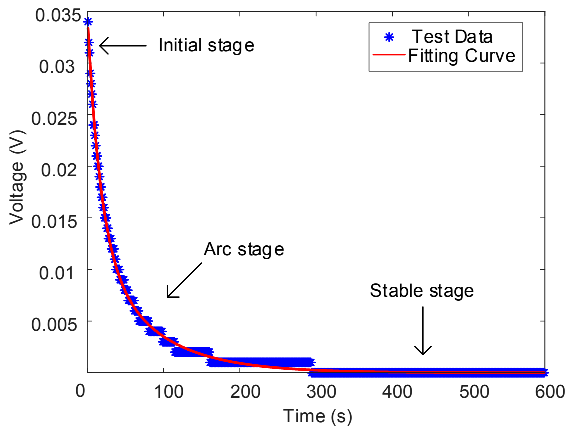

Fig. 4. Ideal fitting result of polarization effects.

III. EEC Model with Additional Resistance

A. Issues on the Result of Fitting

Fig. 5 and 6 are the fitting results of standing 600 s and 40 s, respectively. It can be seen that the longer the t-V curve is fitted, the better the fitting effect of the arc stage, but the poorer the fitting effect in the initial stage for standing 600 s. The fitting for standing 40 s has good effect in the initial and stable stages, but the fitting effect in the arc stage is poor. Since the accuracy of the initial and stable stage is more important, it seems that the traditional way is more reasonable. However, the way is unstable, situation that the stable stage is greatly deviated, which will seriously affect the fitting effect, as shown in Fig. 7. Since fitting results of standing 40 s is accuracy, this situation is difficult to find if not stand for longer periods.

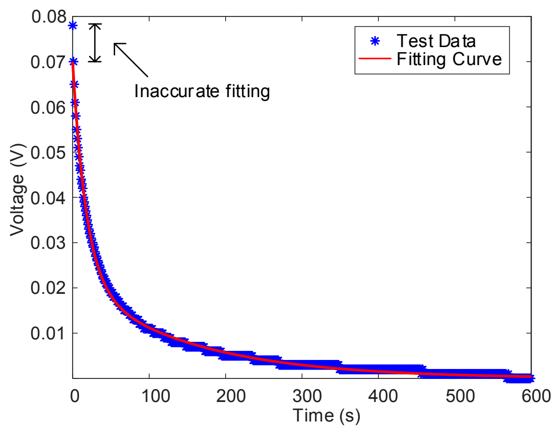

Fig. 5. Fitting result of HPPC with 600 s at 0% SOC.

B. EEC Model with Additional Resistance

Fitting accuracy for standing 600 s is higher than 40 s. In order to solve the poor fitting in initial stage shown in Fig. 5, if a higher order model is used, the added capacitance will be extremely small. Therefore, a series equivalent resistance is used to improve the fitting accuracy without increasing the model order. The improved second-order EEC model is shown in Fig. 8.

The curve with standing 600 s is fitted. Then calculate the additional resistance using the same method as calculating of the DC resistance based on the voltage difference between the initial point and the fitted curve. The advantages of this model will be demonstrated in the next section.

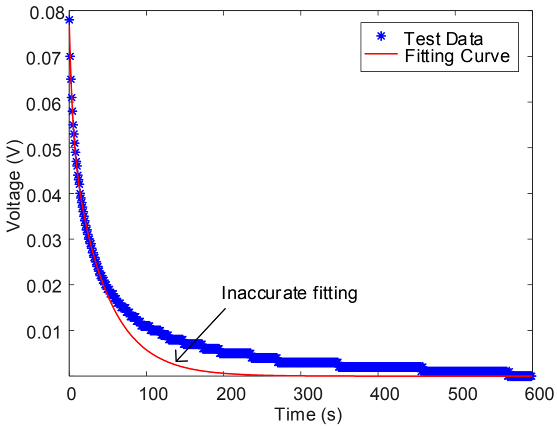

Fig. 6. Fitting result of HPPC with 40 s at 0% SOC.

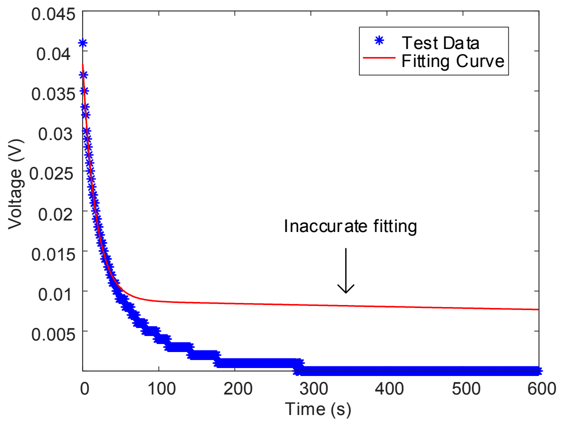

Fig. 7. Issue in fitting result of HPPC with 40 s at 60% SOC.

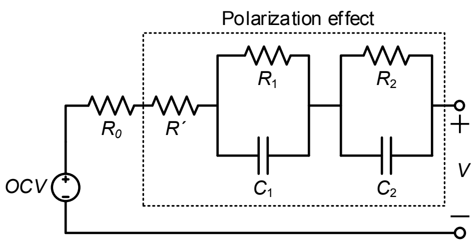

Fig. 8. Second-order EEC model with additional resistance.

IV. Parameters Fitting Based on Genetic Algorithm

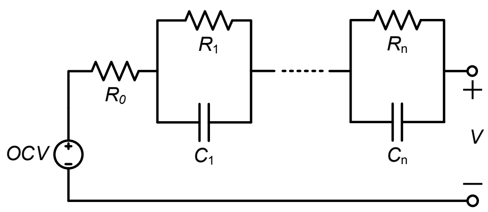

Fig. 9. N-order EEC model.

Not all SOC points need to use high-order models, and the complexity of solving differential equations can be well reduced by appropriately reducing the order of EEC. An adaptive adjustment order EEC model based on genetic algorithm is designed to find the lowest order model that can effectively fit a SOC point.

As shown in Fig. 9, the EEC model starts with the first-order model and increases the order one by one before satisfying the fitting accuracy. Before trying the third-order EEC model, the second-order circuit model with additional resistance is preferentially used. If the fitting accuracy is satisfied, the third-order EEC model will not be used.

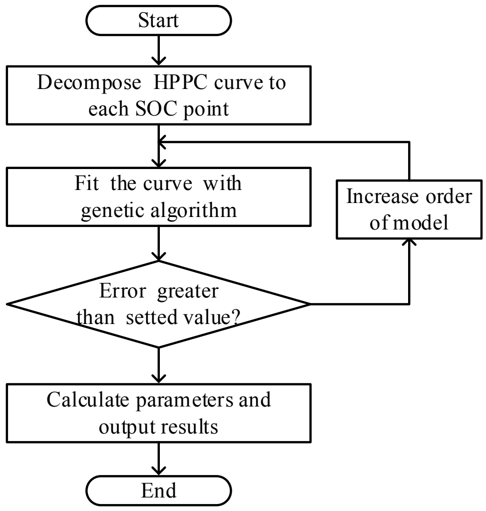

Fig. 10. Flow chart of adaptive EEC model algorithm.

The running process of the entire program is shown in Fig. 10. Each SOC is fitted, and the process is stopped when the converge value of objective function is less than the set value, otherwise the model will be changed and fitting process is repeated. The objective function is as follows

$$\begin{cases} Minimum\ obj: f_1 \\ f_1 = 100\times((v_{real}[1]-v_{fit}[1]))^2 + \sum_{t=2}^{599}(v_{real}[t]-v_{fit}[t])^2 \\ +100\times((v_{real}[600]-v_{fit}[600]))^2 \end{cases} \quad (2)$$

where v_{fit}, v_{real} represent the simulation and experimental data. In order to ensure a good fitting effect in the initial stage and the stable stage. Increase the weighting factor for the initial and the end point.

$$V = OCV + V_{10}e^{-t/\tau_1} + \cdots + V_{n0}e^{-t/\tau_n} \quad (3)$$

The equation of voltage change after open cell in high order EEC models can express by (3). $\tau_1,\cdots,\tau_n$ have similar definition to (1).

Fig. 11 is the fitting result of second-order EEC model with GA. Increasing the weight of first and last point in objective improves fitting accurately in initial and stable stage. Although the arc stage is not fit well but it has little effect on the whole fitting curve than other stages and this method is stable at 60% SOC.

Fig. 12 is the result of GA by different EEC model at 0% SOC. With the order increases, the result of GA becomes smaller. The proposed EEC model have a smaller objective value than basic second-order model. Fig. 13 and 14 are the fitting effect of different EEC model. The proposed model has better fitting accuracy in the arc stage.

Table I lists the applicable EEC models for each SOC point. A, B, C, represent first-order second-order and second-order with additional resistance EEC model respectively. The current used in the HPPC test is 3 A within 0%-80% SOC, 1.5 A at 90% SOC and 0.16 A at 100% SOC.

TABLE I. TYPE OF MODEL APPLY TO EACH SOC

SOC (%)	0	5	10	20	30	40
Type of model	C	C	B	A	A	A
SOC (%)	50	60	70	80	90	100
Type of model	B	A	B	B	A	A

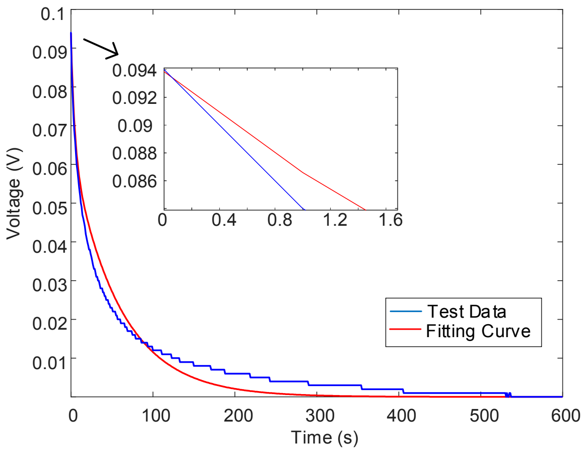

Fig. 11. Fitting result of second-order EEC model by GA at 0% SOC.

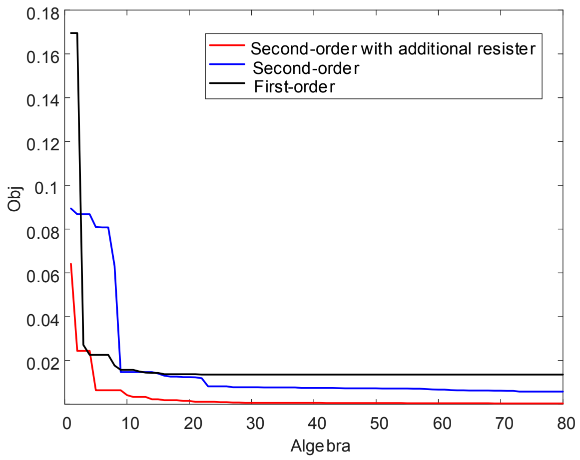

Fig. 12. Convergence of different EEC model by GA at 0% SOC.

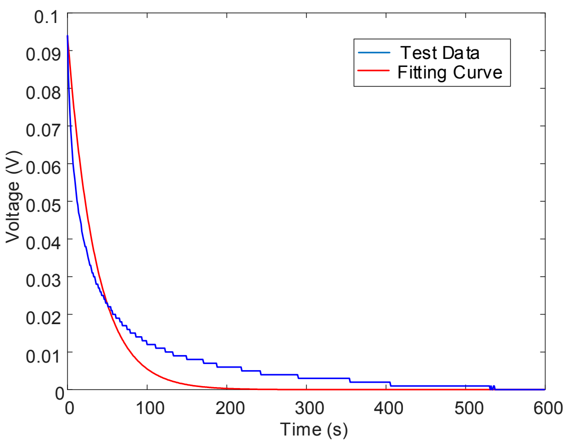

Fig. 13. Fitting result of first-order EEC model by GA at 0% SOC.

Ambient temperature fix at 20° C. If the objective less than 0.0001 when converging, the EEC model is adopted. It can be seen from the results that the most of SOC can be fitted by first-order model and only a few SOC require second-order model, and 0% SOC requires the proposed model.

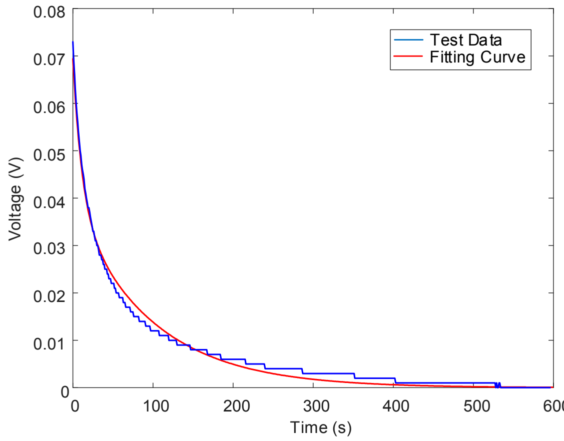

Fig. 14. Fitting result of second-order EEC model with additional resistance by GA at 0% SOC.

V. CONCLUSION

This paper provides an in-depth analysis of parameters fitting of HPPC test. Firstly, by analyzing the issues appear in fitting process, second-order EEC model with additional resistance is proposed, which improves the accuracy of parameters without increasing the model order. Then, a fitting strategy based on GA aims at finding the lowest order of the EEC model under a certain fitting precision is introduced. Finally, Verification experiments are performed at the same SOC by different EEC model, the result shows the obtained model has a better fitting accuracy.

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